7.3 Multiplying and Dividing Rational Expressions

Name _____ Period ____ Date _____

Algebra II 7.3 · Student sheet

I can…

- simplify rational expressions and state the excluded values.
- multiply rational expressions.
- divide rational expressions.

Vocabulary · fill in the missing word

Rational expression — a fraction whose numerator and _____ are polynomials

Simplified form — the numerator and _____ have no common factors other than ±1

Excluded value — a value that makes a _____ 0

Part 1 · Cancel or not?

Is each simplification correct? Write ✓ or ×. Explain each ×.

Simplification	✓ or ×	Why?
$\frac{x+5}{5} = x$		
$\frac{3x}{3y} = \frac{x}{y}$		
$\frac{x^2-4}{x-2} = x+2$		
$\frac{x+2}{x+4} = \frac{2}{4}$		
$\frac{2x+6}{2} = x+3$		
$\frac{x^2+x}{x} = x+1$		
$\frac{x-3}{3-x} = -1$		
$\frac{x}{x+1} = \frac{1}{1}$		

Part 2 · Notes: fill in the blanks

Simplify: _____ first, then divide out common _____ (never terms).

Excluded values come from the _____ denominator.

To multiply: factor, divide out, then _____. To divide: multiply by the _____.

Part 3 · Examples (work them with the class)

Example 1

Simplify $\frac{x^2 - 2x - 15}{x^2 - 9}$. State the excluded values.

Example 2

Find the product: $\frac{8x^3y}{2xy^2} \cdot \frac{7x^4y^3}{4y}$

Example 3

Find the product: $\frac{3x - 3x^2}{x^2 + x - 20} \cdot \frac{x^2 + 4x - 5}{3x}$

Example 4

Find the quotient: $\frac{7x}{2x - 10} \div \frac{x^2 - 6x}{x^2 - 11x + 30}$

Part 4 · Partner practice

Partner 1

Simplify $\frac{x^2 + 7x + 12}{x^2 - 16}$. State the excluded values.

Partner 2

Find the quotient: $\frac{x^2 - 1}{x + 4} \div \frac{x - 1}{x^2 + 4x}$